The empirical recurrence for OEIS A236124, proved by transfer matrix

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Abstract

OEIS A236124 counts arrays over $\{0, \ldots, 3\}$ subject to a condition on the order statistics of every 2×2 subblock, and carries an empirical recurrence of order 40 contributed by R. H. Hardin. It is true, and it is decidable rather than empirical. The condition is local to two consecutive rows, so the count is a walk count on $S = 256$ states and is C-finite; whether the conjectured recurrence annihilates it is then settled by a finite exact computation, with the Cayley–Hamilton theorem bounding the work.

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1 The conjecture

OEIS A236124 is "Number of (n+1)X(1+1) 0..3 arrays colored with the maximum plus the upper median minus the lower median of every 2X2 subblock.". It has offset 1 and begins

$$256, 2782, 30660, 337810, 3723012, 41005528, 451716944, 4976199592, \ldots$$

The entry states, as a conjecture contributed by its author and never marked settled:

> Empirical: a(n) = 366*a(n-3) +5214*a(n-4) +31156*a(n-5) +218596*a(n-6) +873948*a(n-7) +2635751*a(n-8) +5977356*a(n-9) -16093852*a(n-10) -118223502*a(n-11) -524290652*a(n-12) -1690658088*a(n-13) -1592328920*a(n-14) +2665018428*a(n-15) +26560571720*a(n-16) +105162595088*a(n-17) +148035552816*a(n-18) +69121017376*a(n-19) -494484957456*a(n-20) -2549401014912*a(n-21) -3683380168064*a(n-22) -1704351546880*a(n-23) +4457382091776*a(n-24) +26850059834368*a(n-25) +39027657178112*a(n-26) +4608574113792*a(n-27) -39172991209472*a(n-28) -121770582523904*a(n-29) -167682872999936*a(n-30) +44004650811392*a(n-31) +256981414051840*a(n-32) +167599333048320*a(n-33) +6043975286784*a(n-34) -84314060488704*a(n-35) -108348177383424*a(n-36) -43980599328768*a(n-37) +7552163119104*a(n-38) +6707799392256*a(n-39) +782757789696*a(n-40)

As of the "Last modified" line on the live entry (Jul 23 2025 09:07:26, revision 7) this is still recorded as empirical, and nothing on the entry records it as proved.

2 What is being counted

The 2×2 subblocks of an $(n + 1) \times 2$ array form an $n \times 1$ grid. Colour each of them by $\sigma = -b + c + d$, where $a \le b \le c \le d$ are its four entries sorted — its minimum, lower median, upper median and maximum. "Coloured with" is the entry's way of asking for a PROPER colouring of that grid: subblocks that are neighbours, horizontally or vertically, must receive different colours.

That reading is fixed by the entries themselves rather than guessed. Take the upper median over 0..2. A 2×2 array has one subblock and no neighbouring pair, so every one of the $3^4 = 81$ arrays qualifies; a 2×3 array has one horizontal pair and 294 of its 729 arrays qualify; a 3×3 array has two horizontal and two vertical pairs and 722 qualify. Those are the three numbers the corresponding entries publish. Imposing no condition would give 729 at 2×3, and counting the two diagonal pairs as neighbours as well would give 0 at 3×3, so neither is what is meant.

3 The count is a walk count

A subblock's colour is decided by two consecutive rows of the array, and the vertical condition compares two consecutive rows of the subblock grid, so the state is the PAIR of consecutive array rows. A step appends a row: that fixes a new subblock row, which must be properly coloured along itself and must differ from the previous subblock row in every position. An $(n+1)$-row array is a walk of $n-1$ steps, so

$$a(n) \;=\; \iota^{\top} M^{\,n-1} \tau, \qquad \iota = \tau = (1, \ldots, 1)^{\top},$$

over the $S = 256$ pairs whose own subblock row is properly coloured. The alignment of the walk index with the entry's own n was checked against its published terms rather than assumed. In particular a is C-finite.

4 The proof

Lemma 1. *Let $q(t) = t^{40} - \sum_i c_i t^{40-i}$ be the characteristic polynomial of the conjectured recurrence. The residual $a(n) - \sum_i c_i a(n-i)$ equals $\iota^{\top} M^{\,j} q(M) \tau$ for the corresponding walk index j, so the recurrence holds from some point on if and only if $\iota^{\top} M^j q(M) \tau = 0$ for all large j, and it is enough to examine $j < S$.*

Proof. Factor M^j out of $M^{j+40} - \sum_i c_i M^{j+40-i}$. For the bound, M^S is an integer combination of $I, M, \ldots, M^{S-1}$ by Cayley–Hamilton, so the numbers $u_j = \iota^{\top} M^j q(M) \tau$ obey the monic recurrence given by the characteristic polynomial of M; S consecutive zeros force every later one, and the last nonzero u_j pins the threshold exactly. □

Theorem 2.

$$a(n) \;=\; 366\, a(n-3) + 5214\, a(n-4) + 31156\, a(n-5) + 218596\, a(n-6) + 873948\, a(n-7) + 2635751\, a(n-8) + 5977356\, a$$

for every $n > 40$.

Proof. The vector $w = q(M)\tau$ was formed by 40 matrix–vector products in exact integer arithmetic and $\iota^{\top} M^j w$ evaluated for $j = 0, 1, \ldots,$ again exactly; those integers vanish from the index corresponding to $n = 40 + 1$ onwards and the run continued until the working vector was identically zero, which settles every larger j at once. □

5 Verification

Three checks, each independent of the algebra above.

First, the model reproduces all 16 terms the entry publishes, exactly, in integer arithmetic. This is what ties the model to the entry: the digraph is built by reading the entry's English, and a misreading gives different counts.

Second, the conjectured recurrence was evaluated directly on those published terms, with no matrices involved, and holds wherever the proved range and the data overlap.

Third, the annihilation test of Lemma 1 was carried out in exact integer arithmetic on the whole vector rather than on a sampled prefix.

References

[1] The OEIS Foundation, *The On-Line Encyclopedia of Integer Sequences*, https://oeis.org, 2026. Sequence A236124.

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[4] R. A. Horn and C. R. Johnson, *Matrix Analysis*, 2nd ed., Cambridge University Press, 2013.